

Yohan Senadheera

SITE RELIABILITY ENGINEERING INTERN | COMPUTER ENGINEERING UNDERGRADUATE

Software Engineering · Distributed Systems · Networking

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SUMMARY

Computer Engineering undergraduate and Site Reliability Engineering Intern at WSO2 with a proven track record of owning complex features end-to-end. Experienced in translating fuzzy requirements into production-ready architecture, having independently designed a zero-inbound multi-cloud CI/CD platform and coordinated a cross-organization CA-signed PKI migration. Strong in C, Python, and JavaScript, backed by a broad core CS foundation. Combines technical execution with proactive communication and leadership, having reviewed code and designs for 40+ students as an instructor. A highly collaborative team player who thrives on taking ambiguous, from-scratch problems through to secure, documented, and maintainable delivery.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL, Bash, YAML, HCL

Core CS: Data structures & algorithms, Operating systems, Computer networks, Database systems, Software architecture, Distributed systems

Platform: Kubernetes (Namespaces, RBAC), Docker, Helm, Argo Workflows/Events/CD, NATS JetStream, Linux

Cloud & IaC: AWS (EC2, S3, IAM, EKS), Azure (AKS, Entra ID, NSGs, Workload Identity Federation), Terraform

CI/CD, Testing & Security: Azure DevOps YAML, GitOps, GitHub Actions, Git, Unit & Automated testing, OIDC, mTLS, RBAC design, STRIDE threat modelling

Web & Data: React, React Native, Node.js, REST API design, MongoDB, Firebase, Prometheus, Grafana

EXPERIENCE

Site Reliability Engineering Intern

WSO2

Apr 2026 – Present

Colombo, Sri Lanka

Zero-Inbound Multi-Cloud CI/CD Platform

- **Architected from a fuzzy brief:** designed a cloud-agnostic, pull-based central control plane driving independent AWS, Azure, and on-premises data planes establishing full system control over Azure DevOps, ensuring resilience against single-cloud outages, and enabling rapid onboarding of new cloud environments.
- **Enforced strict security boundaries:** implemented stringent network isolation and RBAC permission restrictions replacing shared admin credentials, alongside an mTLS message bus with per-identity ACLs and a self-hosted OIDC issuer to eliminate static secrets.
- **Driving the ongoing migration and documentation:** migration is in progress and also maintaining a STRIDE threat model across several design revisions.

Stack: Argo Workflows/Events/CD, NATS JetStream, Kubernetes, AKS, EKS, k3s, Helm, OIDC, AWS, Azure.

PKI Certificate Automation

- **Automated time-consuming manual operations:** transitioned microservices from a tedious manual self-signing process to an automated CA-signed PKI, significantly improving security and operational efficiency; broke the complex epic into 20+ actionable pull requests across 7 repositories.
- **Designed and proposed the security model:** established cross-organization trust via Workload Identity Federation replacing personal access tokens, enforcing least-privilege identity segmentation so no single compromised credential can execute the signing workflow.
- **Contributed to shared infrastructure:** authored reusable modules merged and released in WSO2's public Terraform library, and navigated the infrastructure through the formal Change Request process.

Stack: Terraform, Azure DevOps YAML, Azure Key Vault, Workload Identity Federation, Azure RBAC, GitHub.

Casual Instructor - Database Systems & Programming Methodology

Dept. of Computer Engineering, University of Peradeniya

Nov 2024 – Aug 2025

Peradeniya, Sri Lanka

- Taught and mentored 40+ students in database systems and C programming, reviewing code and design in labs and one-to-one sessions - giving direct, constructive feedback before implementation.

Trainee - People's Bank

Supported daily branch banking operations, customer records and data entry in a high-volume retail environment.

May 2021 – Feb 2022

PROJECTS

P-E-BO Desk Companion — Third Year Project	Dec 2024 – Jul 2025
Owned the React Native app, Firebase backend and AWS deployment for an AI desk companion with 7-emotion detection at 95% accuracy. IESL Inventor of the Year — Top 10 , 3rd place ICIIS 2025 . <i>React Native, Firebase, AWS, Gemini AI</i> .	
MONA Dashboard — Second Year Project	Apr 2024 – Oct 2024
Developed the production rate tracking page, enabling real-time machine monitoring, bottleneck detection and process optimisation; teammates implemented alerting and historical data visualisation. <i>React.js, MongoDB, JavaScript, Python, SCSS</i> .	
Implementing a Private 5G Network — with SLT Mobitel	Aug 2025 – Present
Implemented an edge UPF on Raspberry Pi with actual 5G RAN to optimise 5G user-plane traffic for a private network deployment; teammates managed RAN virtualisation and core server optimisation. <i>Raspberry Pi, Kubernetes, SDN, Wireshark</i> .	
Smart Home Energy Automation System — Mobile Application	Feb 2025 – Apr 2025
Developed a React Native app for a smart home system using Firebase for real-time synchronisation, alerting and remote device control. <i>React Native (Expo), Firebase, IoT, ESP32, Arduino</i> .	
Cloud-Native Platform with Kubernetes Observability & Recovery	Feb 2026
Designed and built a Kubernetes platform running a containerised API and database with auto-recovery and full observability. <i>Kubernetes, Docker, Prometheus, Grafana, Linux</i> .	

EDUCATION

BSc Eng (Hons) specialising in Computer Engineering	Feb 2022 – Present
<i>University of Peradeniya</i>	<i>Peradeniya, Sri Lanka</i>
• Relevant coursework: Advanced Communication Networks (Networking, SDN, Docker), Network Security, Databases, Software Architecture.	
GCE Advanced Level — Nalanda College, Colombo	2020
Combined Mathematics A, Physics A, Chemistry A, General English A.	
GCE Ordinary Level — Nalanda College, Colombo	2017
9 A passes, awarded for School best results.	
CCNA (Cisco Certified Network Associate) — Cisco Networking Academy	Completed
Completed CCNA 1, 2 and 3: networking fundamentals, routing & switching, and enterprise networking, security & automation.	

LEADERSHIP, TEACHING & COMMUNITY

Editor — Hackers’ Club, University of Peradeniya	Aug 2025 – Jul 2026
Manage technical content and communications for a 200+ member technical community; coordinate events across the committee.	
Team Captain — Shooting Sports Association, University of Peradeniya	Jul 2023 – Oct 2025
Founded air gun shooting as a university sport, captained the team for 2 years, and led it to the 2024 Inter-University Air Rifle Championship title.	
Brand Manager / OC Vice President / Team Leader — AIESEC in Kandy	Feb 2022 – Nov 2024
Led marketing campaigns and coordinated international exchange programmes across cross-functional teams; awarded Best Event Marketer.	
Finance Vice Chair — Career Compass, IEEE Young Professionals Sri Lanka	Apr 2025 – Sep 2025
Managed event finances end to end for the Central Province Organizing Committee: budgeting, allocation, post-budget reporting.	
Senior Member — University Explorers’ Club, University of Peradeniya	Jul 2024 – Present
Organise outdoor expeditions and adventure-based leadership programmes.	

CERTIFICATIONS & AWARDS

IESL Undergraduate Inventor of the Year — Top 10 Finalist, and 3rd Place ICIIS 2025 , for P-E-BO	2025
Coders V11.0 — 9th Place (ACES, Peradeniya) MoraExtreme 9.0 — 27th Place (IEEE, Moratuwa)	Oct 2024
Inter-University Air Rifle Championship — Men’s Team Magam Shooting Sports Colours	Aug 2024
WSO2 Certified Identity Server Practitioner	In Progress
WSO2 Certified Integration Practitioner	In Progress
AWS Technical Essentials — Amazon Web Services	Jan 2026
Docker / Kubernetes for Beginners — KodeKloud	Mar 2026
5G for Everyone — Qualcomm	Sep 2025
Hands-on Linux Commands & Shell Scripting — IBM	May 2025